AFFINE FRAMES AND OUTER-TWISTED DESCENT IN AN EXPLICIT M_{23}-COVER

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Abstract. Let $X \to \mathbf{P}^1_T$ be the explicit rational degree-23 cover with geometric monodromy M_{23} constructed by Huang, Jackson, Lee, Poonen, Pries and Zhang. Above the rational branch point $T = 0$, the fibre has type $1^7 2^8$. Its seven unramified points carry a Fano plane and its eight ramification points, cut out by an irreducible octic R_8, form a set on which $\mathbf{F}_2^3$ acts freely and transitively.

We show that translation classes of ordered affine frames on these eight points form a canonical 168-element torsor. The seven nonzero affine differences are canonically labelled by the seven Fano lines, so after choosing an ordered source basis this torsor is equivariantly identified with the 168 incidence-reversing point–line bijections of the Fano plane. A conjugate copy of the Fano plane meets the original in a noncollinear triple. The marked cover canonically orders this triple; joining each vertex to its successor gives an ordered line basis and hence an affine-frame class without first choosing a point–line bijection. The resulting bijection is denoted by Φ_T.

For $\sigma \in \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$, let w_σ be the normalized sheet permutation extracted from the evaluated Grothendieck–Teichmüller associator. Tame specialization at $T = 0$ proves the identity

$$\rho_{R_8}(\sigma) = \mathrm{pair}(w_{\sigma^{-1}})$$

on the eight ramification points, where pair denotes the permutation induced on the eight two-sheet inertia orbits. Thus GT transport of the ordered triple and Galois transport of the affine-frame torsor of R_8 agree. Equivalently, the graph of Φ_T is preserved after the source action is transported through the independently defined point–line outer action; this is what we call *outer-twisted arithmetic descent*. No such graph is fixed by the ordinary untwisted diagonal action.

The theorem uses the known model over $\mathbf{Q}$. It neither produces a bare Galois-fixed point–line bijection nor independently reproves the rationality of the published Nielsen class. The specialization argument is standard; the contribution is the explicit Mathieu-specific affine-frame realization and its cut-to-correlation factorization.

1. Introduction

1.1. The problem. The construction of [1] starts from the Nielsen class

$$\mathrm{Ni}(2A, 23A, 23B),$$

the set of generating triples (g_1, g_2, g_3) in the indicated conjugacy classes, with $g_1 g_2 g_3 = 1$, modulo simultaneous conjugation by M_{23}. It has seven elements. The absolute Galois group fixes one of these seven classes, represented by the published rational cover. The fixed point is known from the explicit equation, but its relation to the exceptional finite geometry inside M_{23} is not transparent.

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The fibre over the rational $2A$ branch point supplies two natural seven-element systems: the seven unramified points and the seven lines of the Fano plane on them. Their permutation characters agree, but their stabilizers in $\mathrm{PSL}_3(2)$ are nonconjugate. Therefore no $\mathrm{PSL}_3(2)$-equivariant pointwise identification exists. A correlation canonically attached to the marking can exist only if one side is transported through the outer point–line automorphism.

The purpose of this paper is to construct, without using the desired correlation as an input, an affine-frame torsor that realizes the required outer point–line action, and then compare its arithmetic transport with the GT return of the cover.

1.2. Terminology for the branch divisor.

Definition 1.1. Let $U \subset \mathbf{P}^1_{\mathbf{Q}}$ be the largest open subscheme over which $X \to \mathbf{P}^1_{\mathbf{Q}}$ is étale, and put

$$D := \mathbf{P}^1_{\mathbf{Q}} \setminus U.$$

We call the closed complement D the *boundary* of the étale locus. For this cover its geometric points are

$$T = 0, \qquad T = \sqrt{-23}, \qquad T = -\sqrt{-23}.$$

Thus $T = 0$ is a rational *boundary point.* The factor R_8 cuts out the eight ramification points in the fibre above that point, and we use *boundary octic* as shorthand for this ramification octic.

This terminology is standard for the complement of an open locus in algebraic geometry and is modeled on the language of topology. It does not mean the metric or point-set topological boundary of U, and *boundary octic* does not denote another degree-eight cover of the T-line.

1.3. Markings and the cut terminology.

Definition 1.2. A *marking* consists of an identification of the geometric generic fibre with $\{1, \dots, 23\}$, an ordering of the three branch cycles with the involution x first, the tangential direction given by the parameter T at $T = 0$, and a distinguished sheet P. For the published marking, $P = 8$. Transporting the marking by a sheet permutation transports all four pieces of data simultaneously.

Accordingly, *canonical* in this paper means canonical relative to this marked cover and functorial under transport of the marking. It does not mean that a rational origin, basis, or point–line bijection has been chosen.

Definition 1.3. Let y be the marked order-23 branch cycle. Its cyclic ordering of the sheets, together with P, determines the involution c_T that reverses that ordering and fixes P. Conjugating x by c_T gives a second Fano plane. We call it the *twin plane* because c_T exchanges the two isomorphic marked presentations:

$$\mathcal{P}_x^T := \mathrm{Fix}(x^{c_T}), \qquad \mathcal{L}_x := \{\text{lines of the Fano plane on } \mathrm{Fix}(x)\}.$$

Their point sets meet in a noncollinear triple. We call this triple the *cut triangle*: it is the common three-sheet slice along which the two marked presentations are compared, not a topological cut. The marking distinguishes its three vertices and gives a cyclic order $u \to v \to P \to u$. An *oriented cut* means this cyclically ordered triangle, and its *successor-line basis* is

$$\big(\overline{uv}, \overline{vP}, \overline{Pu}\big),$$

where $\overline{ab}$ is the original Fano line through a and b.

Definition 1.4. A *correlation* from the twin plane to the original plane is an incidence-reversing bijection from $\mathcal{P}_x^T$ to $\mathcal{L}_x$. We reserve Φ_T for the correlation determined by the successor-line rule on the oriented cut.

Definition 1.5. By *outer-twisted arithmetic descent* of Φ_T we mean the compatibility proved in corollary 6.3: GT transports the marked source cut, Galois transports the independently constructed affine frame on the roots of R_8, and the cut-to-correlation construction commutes with those two transports. This is weaker than an untwisted fixed graph, which would require one correlation to be invariant when the same $\mathrm{PSL}_3(2)$ element acts diagonally on its source points and target lines.

1.4. **Main results.** Let A be the set of eight ramification points above $T = 0$. The field calculation of [2] gives

$$\mathrm{Gal}(\mathrm{Spl}_{\mathbf{Q}}(R_8)/\mathbf{Q}) \cong 2^3{:}\,\mathrm{PSL}_3(2),$$

acting affinely on A. Let $W \cong \mathbf{F}_2^3$ be its translation subgroup and regard A as a W-torsor, meaning that W acts freely and transitively. An *ordered affine frame* is a quadruple (a_0, a_1, a_2, a_3) for which the three differences $a_i - a_0$ form a basis of W. Define

$$\mathcal{F}(A) = \{\text{ordered affine frames of } A\}/W.$$

Theorem 1.6 (Affine-frame/correlation theorem)**.** *The set $\mathcal{F}(A)$ has 168 elements and carries a free transitive $\mathrm{PSL}_3(2)$-action. Nonzero differences in A are canonically labelled by Fano lines. After an ordered basis of the twin point plane is chosen, this labelling gives an equivariant bijection*

$$\mathcal{F}(A) \xrightarrow{\sim} \mathrm{Corr}(\mathcal{P}_x^T, \mathcal{L}_x).$$

For the published oriented cut, the affine frame is constructed before the correlation, and the resulting correlation is Φ_T.

The Galois comparison is stronger than equality on the 168-element quotient. It already holds on the eight roots of R_8.

The eight transpositions of x give eight two-element $\langle x\rangle$-orbits in the marked generic fibre. Every $c \in C_{M_{23}}(x)$ permutes these orbits; write

$$\mathrm{pair}(c) \in \mathrm{Sym}(A)$$

for the induced permutation after tame specialization identifies the orbits with the roots of R_8. For $\sigma \in \mathfrak{G}_{\mathbf{Q}}$, the element $w_\sigma \in C_{M_{23}}(x)$ is the unique sheet permutation that returns the GT-transported marked branch-cycle tuple to the chosen x-frame; the full construction is given in §5.

Theorem 1.7 (GT–boundary transport theorem)**.** *Use the rational tangential basepoint at $T = 0$, namely the formal direction $T \to 0$. For every $\sigma \in \mathfrak{G}_{\mathbf{Q}}$,*

$$\rho_{R_8}(\sigma) = \mathrm{pair}(w_{\sigma^{-1}}) \quad \textit{in } \mathrm{Sym}(A),$$

where $w_\sigma \in C_{M_{23}}(x)$ is the normalized GT frame return. Consequently the cut-to-frame map is Galois-equivariant and the induced correlation Φ_T has outer-twisted arithmetic descent.

1.5. **What is and is not claimed.** The phrase *outer-twisted descent* is essential. We do not claim that a single point–line bijection is fixed by the ordinary diagonal action. In fact the exact 168-element calculation proves that no such fixed correlation exists. We call such a hypothetical invariant graph *bare* or *untwisted*, because it would use no independently transported point–line outer action. Instead, the source point set is transported through the outer action supplied by the affine-frame torsor of the octic.

We also use the rational equation of [1]. Therefore theorem 1.7 explains why two transports attached to that cover coincide; it is not an equation-free derivation of the rationality of the published Nielsen class. Finally, GT means the actual image of $\mathfrak{G}_{\mathbf{Q}}$ in the profinite Grothendieck–Teichmüller group. No action of arbitrary abstract GT elements on the number field of R_8 is asserted.

2. The $2A$ fibre and its Fano modules

2.1. **The marked Mathieu data.** Fix the published permutation representation of M_{23} and the elements

$$\begin{aligned}
x &= (3\,21)(4\,16)(9\,22)(10\,15)(11\,20)(12\,19)(13\,18)(14\,17),\\
y &= (1\,2\,11\,10\,16\,9\,6\,3\,23\,19\,20\,14\,21\,17\,4\,8\,22\,5\,18\,15\,13\,7\,12),\\
r &= (3\,18)(4\,16)(5\,6)(7\,23)(10\,14)(12\,19)(13\,21)(15\,17).
\end{aligned}$$

Here x is inertia at $T=0$ and y is an order-23 branch cycle. Put $z=(xy)^{-1}$. The involution r is the *branch reflection*: it fixes the $2A$ branch cycle and exchanges the two order-23 branch cycles with the required inversions,

$$x^r = x, \qquad y^r = z^{-1}, \qquad z^r = y^{-1}.$$

Thus the word "reflection" refers to reversing the ordered pair of nonrational branch points. Put

$$C = C_{M_{23}}(x), \qquad K = \ker(C \curvearrowright \mathrm{Fix}(x)).$$

Then

$$|C| = 2688, \qquad K \cong (\mathbf{Z}/2)^4, \qquad C/K \cong \mathrm{PSL}_3(2).$$

The fixed set

$$H = \mathrm{Fix}(x) = \{1,2,5,6,7,8,23\}$$

is the point set of a Fano plane. We order its lines as

$$\begin{aligned}
&L_1 = \{1,2,8\}, \quad L_2 = \{1,5,23\}, \quad L_3 = \{1,6,7\},\\
&L_4 = \{2,5,7\}, \quad L_5 = \{2,6,23\}, \quad L_6 = \{5,6,8\},\\
&L_7 = \{7,8,23\}.
\end{aligned}$$

The quotient $K/\langle x\rangle$ is the irreducible three-dimensional *line module* for $\mathrm{PSL}_3(2)$: its seven nonzero vectors are identified with the seven Fano lines. Dually, the three-dimensional representation whose nonzero vectors are the seven Fano points is the *point module*.

2.2. **The special fibre.** For the integral equation of the cover, specialization at $T=0$ gives

$$F_{\mathrm{int}}(0,V) = 23^4 H_7(V) R_8(V)^2,$$

where H_7 and R_8 are irreducible of degrees 7 and 8. The roots of H_7 are the seven unramified points, while the roots of R_8 are the eight ramification points. In the marked generic fibre these eight points are the transposition orbits of x:

$$\begin{aligned}
&\{3,21\}, \quad \{4,16\}, \quad \{9,22\}, \quad \{10,15\},\\
&\{11,20\}, \quad \{12,19\}, \quad \{13,18\}, \quad \{14,17\}.
\end{aligned}$$

The action of C on these pairs has kernel $\langle x\rangle$ and image

$$C/\langle x\rangle \cong 2^3{:}\,\mathrm{PSL}_3(2).$$

Its regular normal subgroup is

$$W = K/\langle x\rangle \cong \mathbf{F}_2^3.$$

For two distinct ramification points, the two elements of K carrying one to the other differ by x and determine one Fano line. This labels the seven nonzero differences in W by $L_1,\dots,L_7$; each line labels four of the 28 unordered pairs.

Remark 2.1. The phrase *boundary octic* always has the restricted meaning of definition 1.1: the ramification octic in the single fibre above $T = 0$.

3. The point–line correlation torsor

3.1. **The twin plane.** Index the letters along the y-cycle by $\mathbf{F}_{23}$. The orientation-reversing involution of definition 1.3 is

$$a \longmapsto -a + 7$$

lifts to

$$c_T = (1\,3)(2\,6)(4\,22)(5\,17)(7\,19)(9\,11)(10\,16)(12\,23)(13\,20)(14\,15)(18\,21).$$

It fixes the marked point $P = 8$ and reverses the oriented 23-cycle. The twin inertia is x^{c_T} and its fixed set is

$$H^T = \{2, 3, 6, 8, 12, 17, 19\}.$$

The two seven-point sets meet in the noncollinear triple

$$T = H \cap H^T = \{2, 6, 8\}.$$

The branch reflection acts on the original Fano plane as a transvection; its pointwise fixed line, or *axis*, is $\{1, 2, 8\}$. The exact incidence calculation associated with the marking distinguishes 2 from 6, while $P = 8$ is already distinguished. Thus the abstract oriented cut of definition 1.3 is, in the published labels,

$$2 \longrightarrow 6 \longrightarrow 8 \longrightarrow 2.$$

3.2. **Correlations.** As in definition 1.4, write $\mathrm{Corr}(\mathcal{P}_x^T, \mathcal{L}_x)$ for the set of correlations from the twin plane to the original plane. The source point module is the three-dimensional vector space whose seven nonzero vectors are $\mathcal{P}_x^T$; the target line module is the analogous vector space whose nonzero vectors are $\mathcal{L}_x$. Over $\mathbf{F}_2$, projective scalars are trivial, so correlations are precisely the projectivizations of linear isomorphisms between these two modules.

Proposition 3.1. *There are* 168 *correlations. Source collineations act freely and transitively on the left, target collineations act freely and transitively on the right, and the actions commute. Under the untwisted diagonal* $\mathrm{PSL}_3(2)$*-action the orbit sizes are*

$$28, \quad 42, \quad 42, \quad 56,$$

so there is no fixed correlation.

Proof. An invertible linear map between two three-dimensional $\mathbf{F}_2$-spaces is determined by an ordered basis and there are

$$|\,\mathrm{GL}_3(2)| = (2^3 - 1)(2^3 - 2)(2^3 - 2^2) = 168$$

such maps. Left and right composition give the two regular actions. An untwisted fixed correlation would conjugate a point stabilizer to a line stabilizer. Those subgroups have order 24 but are nonconjugate in $\mathrm{PSL}_3(2)$, so no fixed correlation exists. The displayed orbit sizes are an exact enumeration. □

3.3. **The oriented-cut correlation.** Send each cut vertex to the Fano line through that vertex and its successor. On the ordered basis $(2,6,8)$ this gives

$$2 \longmapsto L_5, \qquad 6 \longmapsto L_6, \qquad 8 \longmapsto L_1.$$

Since both triples are projective bases, there is a unique correlation with these three values; denote it by Φ_T.

It is useful to state explicitly what happens when the cut is reversed. Let $\mathcal{P}_x$ be the original point set and $\mathcal{L}_x^T$ the twin line set. Transport by the involution c_T defines the reverse correlation $\Phi_T^- : \mathcal{P}_x \to \mathcal{L}_x^T$ by

$$\Phi_T^-(q^{c_T}) = \Phi_T(q)^{c_T} \qquad (q \in \mathcal{P}_x^T). \tag{1}$$

This formula, together with $c_T^2 = 1$, is the entire compatibility needed between the two orientations. In particular, we do not assert a *polarity*, which ordinarily means an involutive self-correlation of one projective plane; our source and target are the marked twin and original planes.

Proposition 3.2. *The successor-line rule defines a unique correlation Φ_T. In the published labelling its graph is*

$$\begin{array}{r|c} 2 & \{2,6,23\} \\ 6 & \{5,6,8\} \\ 12 & \{1,6,7\} \\ 19 & \{1,5,23\} \end{array} \qquad \begin{array}{r|c} 3 & \{7,8,23\} \\ 8 & \{1,2,8\} \\ 17 & \{2,5,7\} \\ & \end{array}$$

It is independent of simultaneous sheet labels, and reversal of the cut obeys (1)*.*

Proof. The three basis values determine a unique linear isomorphism and therefore a unique projective correlation. Direct incidence calculation gives the displayed graph. The cut, successor operation, line through two points, and projective extension all commute with simultaneous conjugation. Transporting the construction by c_T gives (1); transporting once more returns the original graph because $c_T^2 = 1$. The exact certificate rebuilds the construction on generators of $\mathrm{Sym}(23)$ and verifies this formula. $\square$

4. The affine-frame realization

4.1. **Translation-reduced frames.** An ordered affine frame of A is a quadruple (a_0, a_1, a_2, a_3) for which

$$a_1 - a_0, \quad a_2 - a_0, \quad a_3 - a_0$$

is a basis of W. Quotient frames by simultaneous translation and write

$$\mathcal{F}(A) = \{\text{ordered affine frames of } A\}/W.$$

Proposition 4.1. *There are* 1344 *ordered affine frames and* 168 *translation classes. The affine group* $2^3{:}\,\mathrm{PSL}_3(2)$ *acts on* $\mathcal{F}(A)$ *through its linear quotient, freely and transitively.*

Proof. There are eight origins and, successively, 7, 6, and 4 choices for an ordered basis of differences. Thus there are $8 \cdot 7 \cdot 6 \cdot 4 = 1344$ frames. The translation action is free and has order eight. The quotient is the set of ordered bases of W, on which $\mathrm{GL}_3(2) = \mathrm{PSL}_3(2)$ acts regularly. $\square$

4.2. **Comparison with correlations.** Fix an ordered projective basis $B = (q_1, q_2, q_3)$ of the twin plane; in the Fano plane this means three noncollinear points. Given a frame class, label its three ordered differences by Fano lines $(L_{i_1}, L_{i_2}, L_{i_3})$. There is a unique correlation sending q_j to L_{i_j}.

Theorem 4.2. *For every ordered projective basis B, the preceding construction defines a bijection*

$$\Psi_B : \mathcal{F}(A) \xrightarrow{\sim} \mathrm{Corr}(\mathcal{P}_x^T, \mathcal{L}_x).$$

More precisely, for every $F \in \mathcal{F}(A)$, if g simultaneously transports all the marked data, then

$$\Psi_{B^g}(F^g) = \Psi_B(F)^g. \tag{2}$$

Here the superscript denotes transport of the source points, target lines, and affine frame. With $B = (2, 6, 8)$, exact enumeration verifies (2) on all 2688 elements of C.

Proof. An ordered frame class is an ordered basis of the line module. An ordered source projective basis and an ordered target dual basis determine a unique linear isomorphism, hence a unique correlation. Reversing the construction recovers the line basis, so the map is bijective. Linear extension commutes with transport, which gives (2). The final assertion is the exact finite check. □

4.3. **The cut selects a frame before it selects a correlation.** The successor-line basis of the oriented cut is

$$(L_5, L_6, L_1).$$

Through the canonical difference labelling, it determines a frame class. Using the first ramification point as a displayed origin, the normalized representative has indices

$$[1, 5, 8, 7]$$

relative to the displayed ordering of the eight two-sheet inertia orbits, and consists of the four ramification points

$$\big(\{3, 21\}, \{11, 20\}, \{14, 17\}, \{13, 18\}\big).$$

Theorem 4.3. *The oriented-cut construction factors as*

$$\{\textit{oriented cuts}\} \xrightarrow{\ \kappa\ } \mathcal{F}(A) \xrightarrow{\ \Psi_B\ } \mathrm{Corr}(\mathcal{P}_x^T, \mathcal{L}_x).$$

The map κ is defined without Φ_T, is equivariant for all 2688 elements of C, and

$$\Psi_{(2,6,8)}\big(\kappa(2, 6, 8)\big) = \Phi_T.$$

Proof. The first arrow takes the ordered successor lines and replaces each by the uniquely labelled nonzero translation. This uses only the cut, Fano incidence, and the affine torsor of roots of R_8. Equivariance follows because all three operations commute with C. The equality with Φ_T is checked only after the frame is constructed: both correlations take the ordered source basis $(2, 6, 8)$ to (L_5, L_6, L_1), hence they are equal. □

5. GT frame returns

5.1. **The normalized return.** The three branch values are 0 and the two roots of $T^2 + 23$. Choose the rational tangential basepoint at $T = 0$ defined by the formal direction $T \to 0$. This choice fixes the fibre functor and the orientation used to record branch cycles. The arithmetic action on the geometric fundamental group is described by the cyclotomic character $\lambda_\sigma = \chi(\sigma)$ and the evaluated GT associator $f_\sigma(x, y)$. Here the chosen x-frame is the ordered marked branch-cycle tuple whose first inertia generator is x.

When λ_σ is a square modulo 23, the normalized action in the x-frame sends

$$y \longmapsto (y^{\lambda_\sigma})^{f_\sigma(x,y)}.$$

When λ_σ is a nonsquare, the two conjugate order-23 branch points are exchanged; composing with the branch reflection r returns to the chosen ordering.

Because the cover is defined over $\mathbf{Q}$, every arithmetic transform of the marked branch-cycle tuple is equivalent to the original tuple. There is therefore a unique $w_\sigma \in C$ returning the transformed tuple to the chosen x-frame. Uniqueness follows because the published x-frame orbit has size $|C| = 2688$. Right GT returns compose in the opposite order, so

$$R(\sigma) = w_{\sigma^{-1}}$$

is a homomorphism $\mathfrak{G}_{\mathbf{Q}} \to C$.

Remark 5.1. In double-coset language, existence of w_σ says that the evaluated associator satisfies the finite membership condition determined by λ_σ and the chosen x-frame. Here this is a consequence of the known rational model. We do not infer the rational model from that finite membership condition.

6. Tame specialization and global compatibility

6.1. **Inertia orbits and special points.** Let S be the marked 23-element geometric generic fibre. Let $R = \mathbf{Q}[T]^h_{(T)}$ and normalize $\operatorname{Spec} R$ in the cover. The scheme $\operatorname{Spec} R$ is a henselian trait, meaning the spectrum of a henselian discrete valuation ring; its punctured trait is the complement of its closed point.

Lemma 6.1 (Tame orbit-specialization lemma)**.** *Specialization gives a canonical $\mathfrak{G}_{\mathbf{Q}}$-equivariant bijection*

$$\mathrm{sp} : S/\langle x\rangle \xrightarrow{\sim} \{\textit{geometric points of the normalized fibre over } T = 0\}.$$

Under this bijection, singleton inertia orbits correspond to roots of H_7 and two-element inertia orbits correspond to roots of R_8.

Proof. At a geometric point of ramification index e, tameness and smoothness give a completed local map

$$\overline{\mathbf{Q}}[[T]] \longrightarrow \overline{\mathbf{Q}}[[u]], \qquad T \longmapsto au^e$$

with a a unit. After extracting an e-th root of a, this is $T = u^e$. The e sheets over the punctured trait form one tame inertia orbit and specialize to one point. This produces the bijection. Normalization, reduction, and taking the point below a generic branch commute with base-field automorphisms, proving equivariance. This is the finite-set case of tame specialization; see [3, Exposé XIII, §2.10]. □

6.2. **Identification of the two transports.** Let $d(\sigma)$ be the left Galois descent permutation of the marked generic fibre. The branch-cycle lemma sends x to $x^{\chi(\sigma)} = x$, because x has order two and the cyclotomic character is odd. The arithmetic monodromy is M_{23} and $N_{\mathrm{Sym}(23)}(M_{23}) = M_{23}$, so $d(\sigma) \in C$.

The right GT transport by τ is returned by w_τ. Passing from the right action to the left Galois action gives

$$d(\sigma) = w_{\sigma^{-1}} = R(\sigma).$$

Theorem 6.2. *For every $\sigma \in \mathfrak{G}_{\mathbf{Q}}$,*

$$\rho_{R_8}(\sigma) = \mathrm{pair}\big(w_{\sigma^{-1}}\big) \quad \textit{in } \mathrm{Sym}(A).$$

This equality holds before passage to affine frames and includes the translation subgroup W.

Proof. Let $O \subset S$ be a two-element inertia orbit. By lemma 6.1,

$$\begin{aligned}\sigma\big(\mathrm{sp}(O)\big) &= \mathrm{sp}\big(d(\sigma)O\big)\\ &= \mathrm{sp}\big(w_{\sigma^{-1}}O\big).\end{aligned}$$

The left side is the Galois action on the corresponding root of R_8; the right side is the induced action, denoted pair, on the eight two-element inertia orbits. □

Corollary 6.3 (Outer-twisted descent)**.** *For every $\sigma \in \mathfrak{G}_{\mathbf{Q}}$, the diagram*

$$\begin{array}{ccc} \{\textit{oriented cuts}\} & \xrightarrow{\kappa} & \mathcal{F}(A) \\ \downarrow R(\sigma) & & \downarrow \rho_{\mathcal{F}}(\sigma) \\ \{\textit{oriented cuts}\} & \xrightarrow{\kappa} & \mathcal{F}(A) \end{array}$$

commutes. After applying the explicitly natural comparison Ψ_B of (2), the graph of Φ_T is preserved by the corresponding outer-twisted arithmetic transport.

Proof. Taking affine frames and quotienting by translations are functorial for affine bijections, so theorem 6.2 gives equality of the two actions on $\mathcal{F}(A)$. The map κ is C-equivariant by theorem 4.3. Equation (2), applied while the source basis and target frame are transported together, gives the final statement. □

Remark 6.4. The corollary does not trivialize $\mathcal{F}(A)$. The octic has full affine Galois group, and at $p = 599$ Frobenius is a nonzero translation: both septics split while R_8 factors as four quadratics. There is no preferred rational frame. Likewise, proposition 3.1 forbids a bare untwisted fixed correlation.

7. Interpretation and scope

7.1. **What the theorem explains.** The GT return and the octic Galois representation were introduced from different directions: one from arithmetic transport of a marked branch-cycle tuple, the other from the factorization of a special fibre. Theorem 6.2 shows that they are not merely representations with matching finite statistics. They are the generic- and special-fibre realizations of the same descent datum.

The role of the affine structure on the roots of R_8 is also structural. The seven nonzero translations form the line module, dual to the point module on the seven unramified sheets. Translation-reduced frames turn this dual module into a 168-element torsor, precisely the size required for all point–line correlations. The oriented cut supplies an ordered source basis and an ordered target line basis, producing the correlation without selecting a point of the correlation torsor in advance.

7.2. **What remains outside the theorem.** The theorem starts with the explicit cover over $\mathbf{Q}$. It therefore does not give an independent intrinsic criterion that singles out the published class among the seven Nielsen classes before the rational equation is known. Constructing such a criterion would require analogous data for the six conjugate covers or a universal construction over the degree-seven finite étale Hurwitz algebra parametrizing the seven Nielsen classes.

Nor does the theorem extend the number field of R_8 to an action of every abstract element of $\widehat{GT}$. It concerns the arithmetic subgroup coming from $\mathfrak{G}_{\mathbf{Q}}$. Finally, no analytic assertion about zeta functions or Gaussian multiplicative chaos follows from the descent theorem.

7.3. Limits of the novelty claim. The local normal form $T = u^e$, the branch-cycle lemma, and tame specialization are standard and are not claimed as new. The new content is the explicit combination:

$$\begin{gathered}\text{ramification octic}\\ \Downarrow\\ \text{translation-reduced affine frames}\\ \Downarrow\\ \text{Fano correlation torsor}\\ \Downarrow\\ \text{oriented-cut correlation,}\end{gathered}$$

together with the exact Mathieu-specific identification of all actions and the resulting outer-twisted descent statement.

8. Exact certificates

The finite assertions are verified by exact GAP computations. The arithmetic inputs are inherited from the certificate suite accompanying [2]. The companion targets and outputs are in

https://github.com/research-reflexivity/frontier-math-m23-fixed-point.

Result	Target	Type
Fano incidence and Φ_T	`twin-fano-incidence`	GAP
The two regular torsor actions and no-go theorem	`oriented-cut-hurwitz-torsor`	GAP
Affine frames and the comparison map	`affine-frame-correlation-torsor`	GAP
GT/boundary specialization proof audit	`gt-boundary-transport`	formal+exact
Boundary octic and Frobenius witness	`special-fibre-field-tower`	GAP+PARI

The affine-frame certificate enumerates all 168 frame classes and all 168 correlations, proves that both actions are regular, checks the frame/correlation comparison and cut-to-frame map on all 2688 elements of C, and reconstructs Φ_T only after the affine frame has been constructed. The transport audit records separately the standard specialization lemma, the rational-descent input, the exact finite naturality checks, and the claims that are deliberately excluded.

9. Disclosure

The explicit M_{23}-cover, its rationality, and its branch-cycle description are due to Huang, Jackson, Lee, Poonen, Pries and Zhang [1]. The special-fibre fields and their ramification are from [2]. This paper does not reprove those inputs.

The discovery and drafting process used substantial AI assistance. Every finite assertion used in the main construction has an exact certificate; the standard specialization step is proved separately in lemma 6.1 so that the logical dependence on the rational model is explicit.

References

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